Vidyamandir Classes
Gurukul for IITJEE & Medical Preparation

Illuminati - 2019

Advanced Physics Assignment -2B

Liquids, POM, Thermodynamics, SHM & Waves

SINGLE OPTION CORRECT TYPE

1. A vertical capillary is brought in contact with water surface. The radius of the capillary is r, the angle of contact is θ and surface tension of water is σ. The heat liberated while the water rises in the capillary is:

(A) $\dfrac{\sigma^2 \cos^2\theta}{\rho g}$ **(B)** $\dfrac{\pi\sigma^2}{\rho g \cos^2\theta}$ **(C)** $\dfrac{2\pi\sigma^2 \cos^2\theta}{\rho g}$ **(D)** $\dfrac{2\pi\sigma^2}{\rho g \cos\theta}$

2. A string of length 'l' is fixed at both ends. It is vibrating in its 3rd overtone with maximum amplitude 'a'. The amplitude at a distance $l/3$from one end is:

(A) a **(B)** 0 **(C)** $\dfrac{\sqrt{3}a}{2}$ **(D)** $\dfrac{a}{2}$

3. A liquid of volumetric thermal expansion coefficient γ and bulk modulus B is filled in a spherical tank of negligible heat expansion coefficient. Its radius is R and wall thickness is $`t'\,(t << R)$. When the temperature of the liquid is raised by θ, the tensile stress developed in the walls of the tank is:

(A) $B\gamma\,\theta R/2t$ **(B)** $B\gamma\,\theta R/t$ **(C)** $2B\gamma\,\theta R/t$ **(D)** $B\gamma\,\theta R/4t$

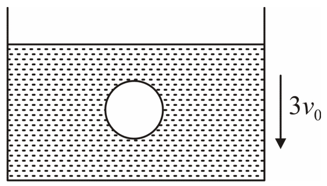
4. Which of the following is greatest in SHM ? (Assuming potential energy = 0 at mean position)

(A) Average kinetic energy with respect to space
(B) Average potential energy with respect to space
(C) Average kinetic energy with respect to time
(D) Average potential energy with respect to time

5. Mean free path of oxygen molecules at 0°C and pressure p is $\lambda_1 = 9.5 \times 10^{-8}$ m. Keeping temperature constant pressure is reduced 100 times, then new mean free path is :

(A) 9.5×10^{-8} m **(B)** 9.5×10^{-10} m **(C)** 9.5×10^{-6} m **(D)** 9.5×10^{-12} m

6. A container filled with viscous liquid is moving vertically downwards with constant speed $3v_0$. At the instant shown, a sphere of radius r moving vertically downwards, in liquid, has a speed v_0. The coefficient of viscosity is η. There is no relative motion between the liquid and the container. Then at the shown instant, the magnitude of viscous force acting on sphere is :



(A) $6\pi\eta r v_0$ **(B)** $12\pi\eta r v_0$ **(C)** $18\pi\eta r v_0$ **(D)** $24\pi\eta r v_0$

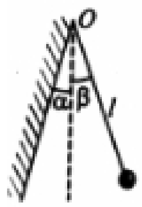
7. A solid sphere and a hallow sphere of the same material and size are heated to the same temperature and allowed to cool in the same surroundings. If the temperature difference between the surroundings and each sphere is T, then:

(A) The hollow sphere will cool at a faster rate for all values of T
(B) The solid sphere will cool at a faster rate for all values of T
(C) Both spheres will cool at the same rate for all values of T
(D) Both spheres will cool at the same rate only for small values of T

8. What is the fraction of molecule below an altitude h in atmosphere ? Assume uniform gravitational field, isothermal conditions, mass of a molecule m, Boltzman constant k, temperature T.

(A) $f = e^{(mgh/kT)}$ **(B)** $f = e^{-(mgh/kT)}$ **(C)** $f = 1 - e^{-(mgh/kT)}$ **(D)** $f = 1 - e^{(mgh/kT)}$

9. A small ball is suspended by a thread of length l = 1m at the point O on the wall, forming a small angle $\alpha = 2°$ with the vertical (as shown in figure). Then the thread with ball was deviated through a small angle $\beta = 4°$ and set free. Assuming the collision of the ball against the wall to be perfectly elastic, find the oscillation period of such a pendulum. (Take $g = \pi^2$)

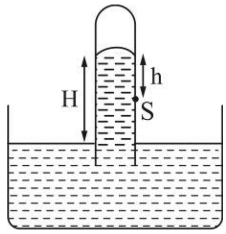


(A) $\frac{2}{3}$sec **(B)** $\frac{4}{3}$sec **(C)** 2 sec **(D)** None of these

10. Two waves are propagating to the point P along a straight line produced by two sources A and B of simple harmonic sound of equal frequency. The amplitude of each wave at P is 'a' and the phase of A is ahead by $\pi/3$ than that of B and the distance AP is greater than BP by 50 cm. Then the resultant amplitude at the point P will be, if the wavelength is 1 meter:

(A) $2a$ **(B)** $a\sqrt{3}$ **(C)** $a\sqrt{2}$ **(D)** a

11. In a mercury barometer used to measure atmospheric pressure, mercury rises to a height H. Now if a small hole is made at point S as shown, mercury comes out from this hole with a speed equal to :



(A) $\sqrt{2gh}$ **(B)** $\sqrt{2g(H-h)}$
(C) $\sqrt{2gH}$ **(D)** None of these

12. Two spheres of same metal have the same volume. But one is solid and the other is hollow. When the change in temperature of both of them is same, which of the following statements about the change in their diameters is true?

(A) More for solid sphere **(B)** More for hollow sphere
(C) Same for both spheres **(D)** It cannot be predicted

13. 1 cm^3 of water absorbs 540 cal of heat to become steam with a volume of 1671 cm^3. If atmospheric pressure is 1.013×10^5 Nm^{-2} and 1 cal = 4.19 J, the energy spent in this process in over coming molecular attraction is :

(A) 540 cal **(B)** 40 cal **(C)** 500 cal **(D)** zero

14. An accurate pendulum clock is mounted on the ground floor of a high building. How much time will it lose or gain in one day if it is transferred to top storey of a building which is h =200 m higher than the ground floor. Radius of earth is $6.4 \times 10^6 m$:

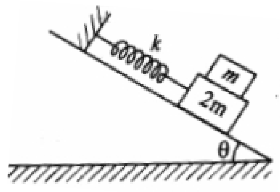
(A) It will lose 6.2 s **(B)** It will lose 2.7 s **(C)** It will gain 5.2 s **(D)** It will gain 1.6 s

15. A standing wave pattern is formed on a string. One of the waves is given by equation $y_1 = a\cos(\omega t - kx + \pi/3)$. Then find the equation of the other wave such that at $x = 0$ a node is formed.

(A) $y_2 = a\sin\left(\omega t + kx + \frac{\pi}{3}\right)$ **(B)** $y_2 = a\cos\left(\omega t + kx + \frac{\pi}{3}\right)$

(C) $y_2 = a\cos\left(\omega t + kx + \frac{2\pi}{3}\right)$ **(D)** $y_2 = a\cos\left(\omega t + kx + \frac{4\pi}{3}\right)$

16. The coefficient of friction between block of mass m and $2m$ is $\mu = 2\tan\theta$. There is no friction between block of mass $2m$ and inclined plane. The maximum amplitude of two block system for which there is no relative motion between both the blocks is :



(A) $g\sin\theta\sqrt{\frac{k}{m}}$ **(B)** $\frac{mg\sin\theta}{k}$ **(C)** $\frac{3mg\sin\theta}{k}$ **(D)** None of these

17. Two similar sonometer wires have fundamental frequencies of 500 Hz. These have same tensions. By what amount the tension should be increased in one wire so that the two wires produce 5 beats/sec:

(A) 1% **(B)** 2% **(C)** 3% **(D)** 4%

18. A new scale N of temperature is divided in such a way that the freezing point of water is $100°N$ and boiling point is $400°N$. The Celsius and new temperature scale readings will be same at :

(A) $-40°C$ **(B)** $40°C$ **(C)** $-50°C$ **(D)** $50°C$

19. A vertical glass jar is filled with water at 10°C. It has one thermometer at the tip and another at the bottom. The central region of the jar is gradually cooled. It is found that the bottom thermometer reads 4°C earlier than the top thermometer. And the top thermometer reads 0°C earlier than the bottom thermometer. This happens because :

(A) The top thermometer is faulty **(B)** The bottom thermometer is faulty

(C) Density of water is maximum at 4°C **(D)** Density of water is minimum at 0°C

20. Two identical heaters are coated with paint. In 1st case $e_1 = 1.0$ and in 2nd case $e_2 = 0.5$. Both are kept in identical chambers which are in similar outside surroundings. If the heaters are switched on, in steady state 1st heater has temperature T_1 on surface and θ_1 of its chamber. 2nd heater has temperature T_2 on surface and θ_2 of its chamber. Then :

(A) $\theta_1 = \theta_2$; $T_1 < T_2$ **(B)** $\theta_1 > \theta_2$; $T_1 = T_2$ **(C)** $\theta_1 = \theta_2$; $T_1 > T_2$ **(D)** $\theta_1 < \theta_2$; $T_1 = T_2$

ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

21. Two holes with area $a = 0.2\text{ cm}^2$, each are drilled one above the other in the wall of a vertical vessel filled with water. The distance between the holes is $H = 50\text{ cm}$. Every second $Q = 140\text{ cm}^3$ of water is poured into the vessel. The streams flowing out of the holes intersect at a point. x is the horizontal distance of this point from the vessel and y is the vertical distance of this point from the water surface in the tank. x and y remains same as the time passes. Choose the correct option(s).

(A) $x = 120$ cm **(B)** $x = 60$ cm **(C)** $y = 75$ cm **(D)** $y = 130$ cm

22. A metallic wire of length L is held between two rigid supports with no initial stress. If the wire is cooled through a temperature θ, then fundamental frequency of oscillation is proportional to : (where, Y = Young's modulus of elasticity of wire, $\rho =$ density, $\alpha =$ thermal coefficient of linear expansion)

(A) $\frac{1}{L}$ **(B)** $\sqrt{Y}$ **(C)** $\sqrt{\alpha / \rho}$ **(D)** $\sqrt{\theta}$

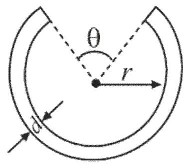
23. A spring of spring constant K is fixed to the ceiling of a lift. The other end of the spring is attached to a block of mass m. The mass is in equilibrium. Now the lift accelerates downwards with an acceleration $2g$. Then choose the correct option(s).

(A) The block will not perform SHM and it will stick to the ceiling
(B) The block will perform SHM with time period $2\pi\sqrt{m / K}$
(C) The amplitude of the block will be 2 mg/K if it performs SHM
(D) The minimum potential energy of the spring during the motion of the block will be 0

24. A certain pure metal has a specific heat of 230 $Jkg^{-1}K^{-1}$ at high temperatures. Which of these are correct?

(A) Molar specific heat of metal is 24.942 $J/mol\text{-}K$
(B) Molar specific heat of metal remains constant for a slight temperature change
(C) Molecular weight of metal is 108 gram
(D) All atoms in metal vibrate at same frequency

25. A thin cylindrical metal rod is bent into a ring with a small gap as shown in figure. On heating the system:



(A) θ decreases, r and d increases **(B)** r and d increases
(C) θ, d and r increases **(D)** θ is constant

26. A mercury barometer is used to measure air pressure in an elevator. If the elevator is stationary, the reading of the barometer is 76 cm. Assume air pressure to remain unchanged, choose correct statements.

(A) If the elevator is accelerated upwards, the reading of barometer is more than 76 cm
(B) If the elevator is accelerated upwards, the reading of barometer is less than 76 cm
(C) If the elevator is moving upwards with constant velocity, the reading of barometer is 76 cm
(D) If the elevator is accelerated downwards with acceleration less than g, the reading of barometer is more than 76 cm

27. A light container having a diatomic gas enclosed within, is moving with velocity v. Mass of gas is m, number of moles is n. Then, choose the correct options.

(A) KE of gas with respect to COM of system is $\frac{3}{2}nRT$
(B) KE of gas with respect to COM of system is $\frac{5}{2}nRT$
(C) KE of gas with respect to ground is $\frac{5}{2}nRT$
(D) KE of gas with respect to ground is greater than $\frac{5}{2}nRT$

28. Two sounding bodies are producing progressive sound wave given by $y_1 = 4\sin(400\pi t)$ and $y_2 = 3\sin(404\,\pi t)$, where t is in second. The two waves superpose near the ears of a person. The person will hear:

(A) 2 beats per second
(B) intensity ratio 7 between maxima and minima
(C) intensity ratio 49 between maxima and minima
(D) 4 beats per second

29. During melting of a slab of ice at 273 K and at atmospheric pressure, which of these are correct ?

(A) positive work is done on ice-water system by atmosphere
(B) positive work is done on atmosphere by ice-water system
(C) internal energy of the ice-water system increases
(D) internal energy of atmosphere increases

30. Initially spring is compressed by x_0 and blocks are in contact. When system is released, then blocks starts moving and after some time contact between blocks breaks, then:

(A) blocks will separate at natural length of spring
(B) after separation, block A performs SHM of amplitude $x_0\sqrt{\dfrac{M_1}{M_1+M_2}}$
(C) after separation, maximum velocity of block A is $x_0\sqrt{\dfrac{k}{M_1+M_2}}$
(D) after separation, block A will perform SHM of amplitude x_0

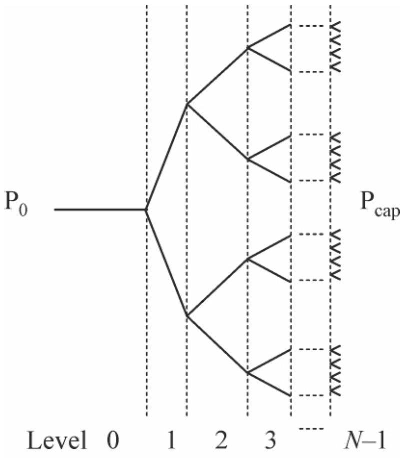
PARAGRAPH TYPE

PARAGRAPH FOR QUESTIONS 31 - 32

Blood vessels are approximately cylindrical in shape, and it is known that for a steady, non-turbulent flow of an incompressible fluid in a rigid cylinder, the difference in pressure of the fluid at the two ends of the cylinder is given by : $\Delta P = \dfrac{8\ell\eta}{\pi r^4}Q$

Where ℓ and r are the length and radius of the cylinder, η is the fluid viscosity and Q is the volumetric flow rate, i.e. the fluid volume that passes the cylinder cross-section per unit time. This expression is often able to provide the correct order of magnitude for the pressure difference in a blood vessel. Moreover, this expression has the same form as Ohm's law, with the volumetric flow rate being interpreted as a current, the difference in pressure as a voltage, and the factor $R = \dfrac{8\ell\eta}{\pi r^4}$ as a resistance.

Consider for example the symmetrical network of arterioles (small arteries) depicted in figure that delivers blood to the capillary bed of a tissue. In this network, at each bifurcation a vessel is divided in two identical vessels. However, the vessels of higher levels are thinner and shorter : Considered that the radii and lengths of vessels in two consecutive levels i and $i + 1$, are related by $r_{i+1} = r_i / 2^{1/3}$ and $\ell_{i+1} = \ell_i / 2^{1/3}$.



Network of arterioles

31. Obtain an expression for the volumetric flow rate, Q_i, in a vessel at any level i, as a function of the total number of levels N, of the viscosity η, of the radius r_0 and length ℓ_0 of the arteriole at level 0, and of the difference $\Delta P = P_0 - P_{cap}$ between the inlet pressure at the arteriole at level $0\left(P_0\right)$ and the pressure at the capillary bed $\left(P_{cap}\right)$.

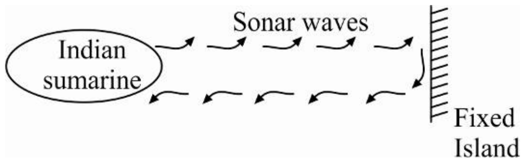
(A) $\Delta P \dfrac{\pi r_0^4}{8N\ell_0\eta}$ **(B)** $\Delta P \dfrac{\pi r_0^4}{2^{i+3}N\ell_0\eta}$ **(C)** $\Delta P \dfrac{\pi r_0^4}{2^{i+3}\ell_0\eta}$ **(D)** $\Delta P \dfrac{\pi r_0^4}{2^{i}N\ell_0\eta}$

32. Calculate the numerical value of the volumetric flow rate Q_0 of the arteriole at level 0, if its radius is $6.0\times10^{-5}m$ and its length is $2.0\times10^{-3}m$. Consider that the pressure at the arteriole inlet is $55 mmHg$ and the vessel network has N = 6 levels linking this arteriole to the capillary bed at the pressure 30 *mmHg*. Consider that the blood viscosity is $\eta = 3.5\times10^{-3}\,kg\,m^{-1}s^{-1}$. Express your result in $m\ell/h$. (Normal atmospheric pressure, $P_0 = 1.013\times10^5\ Pa = 760\ mmHg$)

(A) $15\,m\ell/hr$ **(B)** $4\,m\ell/hr$ **(C)** $1.5\,m\ell/hr$ **(D)** $150\,m\ell/hr$

PARAGRAPH FOR QUESTIONS 33 - 35

An Indian submarine is moving in "Arabian Sea" with a constant velocity. To detect enemy it sends out sonar waves which travel with velocity 1050 m/s in water. Initially the waves are getting reflected from a fixed island and the reflected waves are coming back to submarine.



The frequency of reflected waves are detected by the submarine and found to be 10% greater than the sent waves.

Now an enemy ship comes in front, due to which the frequency of reflected waves detected by submarine becomes 21% greater than the sent waves.

33. The speed of Indian submarine is :

(A) 10 m/sec **(B)** 50 m/sec **(C)** 100 m/sec **(D)** 20 m/sec

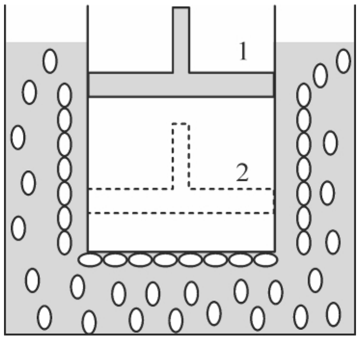
34. The velocity of enemy ship should be:

(A) 50 m/sec. toward Indian submarine **(B)** 50 m/sec. away from Indian submarine
(C) 100 m/sec. toward Indian submarine **(D)** 100 m/sec. away from Indian submarine

35. If the wavelength received by enemy ship is λ' and wavelength of reflected waves received by submarine is λ'' then $\left(\dfrac{\lambda'}{\lambda''}\right)$ equals :

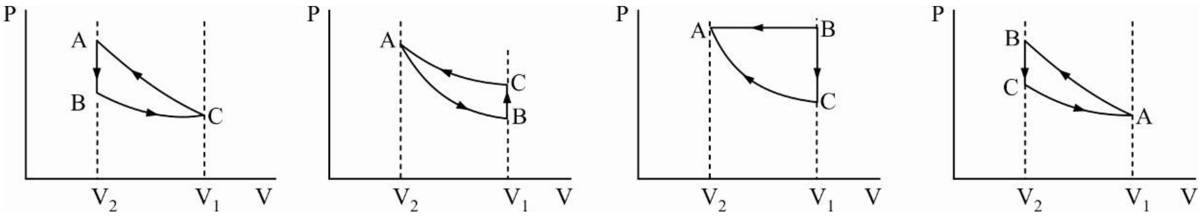
(A) 1 **(B)** 1.1 **(C)** 1.2 **(D)** 2

PARAGRAPH FOR QUESTIONS 36 - 37



A cylinder containing an ideal gas (see diagram) and closed by a movable piston is submerged in an ice-water mixture. The piston is quickly pushed down from position (1) to position (2) (process AB). The piston is held at position (2) until the gas is again at $0°C$ (process BC). Then the piston is slowly raised back to position (1) (process CA).

36. Which of the following P-V diagrams will correctly represent the process AB, BC and CA and the cycle $ABCA$?



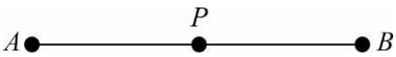
(A) **(B)** **(C)** **(D)**

37. If 100 gm of ice is melted during the cycle ABCA, how much work is done on the gas?

(A) 8 kcal **(B)** 5 kcal **(C)** 2.1 kJ **(D)** 4.2 kJ

PARAGRAPH FOR QUESTIONS 38 - 40

A and B are two fixed points at a distance 3ℓ apart. A particle of mass m placed at a point P experiences the force $2(mg/\ell)\overline{PA}$ and the force $(mg/\ell)\overline{PB}$ simultaneously. Initially at $t = 0$, the particle is projected from A towards B with speed $3\sqrt{g\ell}$.



38. The particle moves simple harmonically with period T and amplitude A :

(A) $A = 2\ell,\ T = 2\pi\sqrt{\dfrac{\ell}{g}}$ **(B)** $A = 3\ell,\ T = 2\pi\sqrt{\dfrac{\ell}{2g}}$

~~**(C)**~~ $A = 2\ell,\ T = 2\pi\sqrt{\dfrac{\ell}{3g}}$ **(D)** $A = \ell,\ T = 2\pi\sqrt{\dfrac{\ell}{g}}$

39. The instant 't' at which the particle arrives at B in terms of the periodic time T will be :

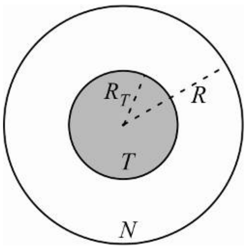
(A) $t = \dfrac{T}{2}$ ~~**(B)**~~ $t = \dfrac{T}{3}$ **(C)** $t = \dfrac{T}{4}$ **(D)** $t = \dfrac{2T}{3}$

40. The velocity of the particle when it reaches B will be :

~~**(A)**~~ zero **(B)** $3\sqrt{g\ell}$ **(C)** $2\sqrt{g}$ **(D)** $g\ell$

PARAGRAPH FOR QUESTIONS 41 - 42

Figure shows a simplified model of a tumour. The tumour is assumed to be spherical with radius R_T surrounded by normal cells.



Hyperthermia is sometimes used together with chemotherapy and radiotherapy in the treatment of cancer. In hyperthermia the cancer cells are selectively heated from the normal human body temperature, $37°C$ to temperatures above $43°C$, inducing their death. Researchers are currently developing carbon nanotubes covered with special proteins capable of binding to tumour cells. When the tissue is irradiated with near-infrared radiation, the nanotubes absorb it in a much greater extent than the surrounding tissues and therefore can be selectively heated as well as the tumour cells to which they are attached.

Consider that the tumour, the normal cells and the surrounding tissue have a constant thermal conductivity *k*, i.e. in the geometry of this problem, the energy that crosses a spherical surface of radius *r* per unit time and per unit area is equal to *k* times the derivative of the temperature with respect to *r*. The nanotubes are uniformly distributed in the tumour volume and are able to deliver a power *P* of thermal energy per unit volume. Assume that the temperature is equal to the normal human body temperature very far away from the tumour.

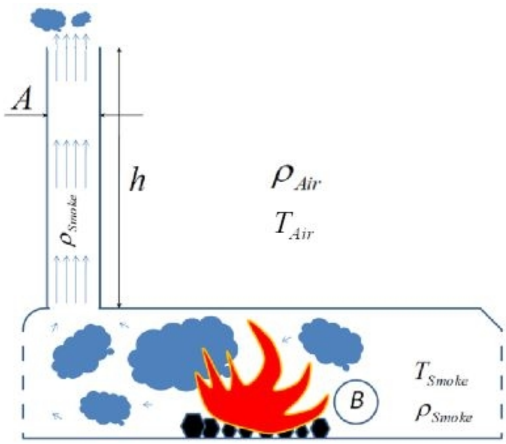
41. Obtain, for the stationary state, how much higher will the temperature at the centre of the tumour become as compared to the normal body temperature ?

(A) $\dfrac{PR_T^2}{3k}$ **(B)** $\dfrac{PR_T^2}{2k}$ **(C)** $\dfrac{PR_T^2}{6k}$ **(D)** $\dfrac{PR_T^2}{k}$

42. Obtain, the minimum power per unit volume, $P_{\min}$, needed to heat up all turmour cells in a tumour with 5.0 *cm* radius to a temperature larger than $43.0°C$. Take the thermal conductivity of the tissue to be equal to $k = 0.60\ WK^{-1}\ m^{-1}$.

(A) $4.3\ kW/m^3$ **(B)** $2.9\ kW/m^3$ **(C)** $1.4\ kW/m^3$ **(D)** $8.6\ kW/m^3$

PARAGRAPH FOR QUESTIONS 43 - 45



Gaseous product of burning are released into the atmosphere of temperature T_{Air} through a high chimney of cross-section *A* and height *h* (see figure). The solid matter is burned in the furnace which is at temperature T_{smoke}. The volume of gases produced per unit time in the furnace is *B*.

Assume that :

- The velocity of the gases in the furnace in negligibly small.
- The density of the gases (smoke) does not differ from that of the air at the same temperature and pressure; while in furnace, the gases can be treated as ideal.

- The pressure of the air changes with height in accordance with the hydrostatic law ; the change of the density of the air (ρ_{air}) with height is negligible. Pressure inside the furnace is same as outside.
- The flow of gases fulfills the Bernoulli equation which states that the following quantity is conserved in all points of the flow : $\frac{1}{2}\rho v^2(z)+\rho gz+p(z)=\text{const}$.

  where ρ is the density of the gas, $v(z)$ is its velocity, $p(z)$ is pressure, and z is the height .
- The change of the density of the gas $\left(\rho_{gas}\right)$ is negligible throughout the chimney.

43. What is the minimal height of the chimney needed in order that the chimney functions efficiently, so that it can release all of the produced gas into the atmosphere? Express your result in terms of $B, A, g, \rho_{air}, \rho_{smoke}$.

(A) $\frac{B^2\rho_{smoke}}{2A^2g\left(\rho_{air}-\rho_{smoke}\right)}$ **(B)** $\frac{B^2\rho_{air}}{2A^2g\left(\rho_{air}-\rho_{smoke}\right)}$

(C) $\frac{B^2}{2A^2g}$ **(D)** $\frac{B^2}{2A^2g}\frac{\rho_{smoke}}{\rho_{air}}$

44. Assume that two chimneys are built to serve exactly the same purpose. Their cross-sections are identical, but are designed to work in different parts of the world : One in cold regions, designed to work at an average atmospheric temperature of $-30°C$ and the other in warm regions, designed to work at an average atmospheric temperature $30°C$. The temperature of the furnace is $400°C$. It was calculated that the height of the chimney designed to work in cold regions is $100m$. How high is the other chimney?

(A) 116 *m* **(B)** 200 *m* **(C)** 145 *m* **(D)** 50 *m*

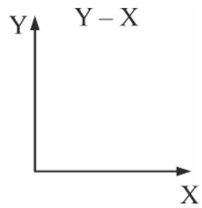
45. How does the pressure of the gases vary with height (z) inside the chimney? Here $P(0)$ is the pressure inside the furnace and h is the minimal height as calculated above.

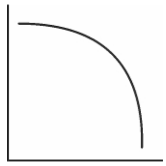
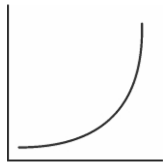
(A) $P(z)=P(0)-\rho_{smoke}gz$ **(B)** $P(z)=P(0)-\left(\rho_{air}-\rho_{smoke}\right)gh-\rho_{smoke}gz$

(C) $P(z)=P(0)-\left(\rho_{air}-\rho_{smoke}\right)gz$ **(D)** $P(z)=P(0)-\rho_{air}gh-\rho_{smoke}gz$

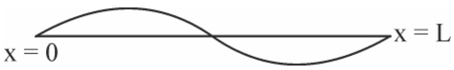
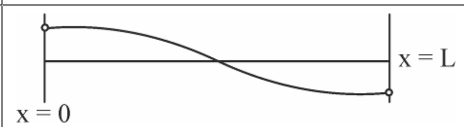
MATCH THE COLUMN TYPE

46. For an adiabatic process, match the graphs given in column II corresponding to the pair given in column I taking the first quantity on Y-axis and second quantity on X-axis.

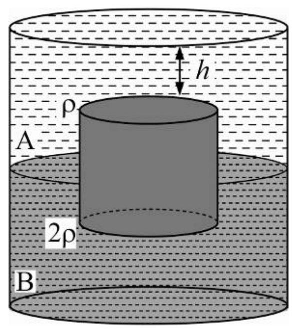


	Column I		Column II
(A)	Pressure-Temperature	**(P)**	
(B)	Volume-Temperature	**(Q)**	
(C)	Pressure-Volume	**(R)**	
(D)	Pressure-Internal Energy	**(S)**	

47. Column I shows different sets of standing waves in a string of length L whose ends are fixed or free according to respective figure and Column-II shows possible equations for them where symbols have usual meaning.

	Column I		Column II
(A)		**(P)**	$y = A\sin\dfrac{\pi x}{L}\cos\omega t$
(B)		**(Q)**	$y = A\sin\dfrac{2\pi x}{L}\cos\omega t$
(C)		**(R)**	$y = A\cos\dfrac{\pi x}{2L}\cos\omega t$
(D)		**(S)**	$y = A\cos\dfrac{\pi x}{L}\cos\omega t$

48. Shown below is a vessel containing liquids A and B of densities ρ and 2ρ respectively. A cylinder of height $2h$ floats with equal volumes submerged in both liquids. The top surface of cylinder is at a depth h below the surface of liquid A. Neglecting atmospheric pressure, match the quantities mentioned in column I with corresponding expression in column II.



	Column I		Column II
(A)	Net force exerted by liquid A on the cylinder	**(P)**	$9\rho gRh^2$
(B)	Net force exerted by liquid B on the cylinder	**(Q)**	$\pi\rho gR^2h$
(C)	Net force exerted by liquids A and B on the left half of the curved part of cylinder	**(R)**	$4\pi\rho gR^2h$
(D)	Net force exerted by liquids A and B on the cylinder	**(S)**	$3\pi\rho gR^2h$
		(T)	$2\pi\rho gR^2h$

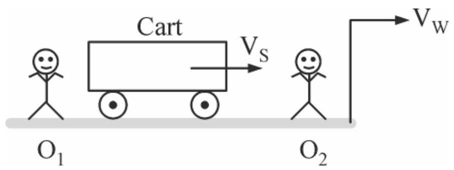
49. Two particle 'A' and 'B' start SHM at $t = 0$. Their positions as a function of time are given by:

$$x_A = A\sin\omega t$$

$$x_B = A\sin(\omega t + \pi/3)$$

	Column I		Column II
(A)	Minimum time when x is same	**(P)**	$\frac{5\pi}{6\omega}$
(B)	Minimum time when velocity is same	**(Q)**	$\frac{\pi}{3\omega}$
(C)	Minimum time after which $v_A < 0$ and $v_B < 0$	**(R)**	$\frac{\pi}{\omega}$
(D)	Minimum time after which $x_A < 0$ and $x_B < 0$	**(S)**	$\frac{\pi}{2\omega}$

50. Observers O_1 and O_2 are at rest and the wall is moving with velocity V_w. Cart is moving with constant velocity V_s towards wall. The source of sound is in the cart, the original frequency of the wave is f. Sound has velocity C w.r.t. ground (medium is stationary). Then match the column-I with column II.



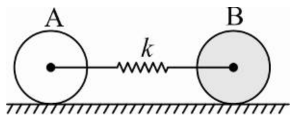
	Column I		Column II
(A)	Wavelength received by O_1 directly form cart	**(P)**	$\left(\frac{C-V_w}{C+V_w}\right)\frac{(C-V_s)}{f}$
(B)	Wavelength received by O_2 directly from cart	**(Q)**	$\frac{C+V_s}{f}$
(C)	Wavelength received by driver of the cart after reflection from wall.	**(R)**	$\left(\frac{C+V_w}{C-V_w}\right)\frac{(C-V_s)}{f}$
(D)	Wavelength received by O_1 after reflection from wall	**(S)**	$\frac{C-V_s}{f}$
		(T)	None of these

NUMERICAL/SUBJECTIVE TYPE

51. The time period of a physical pendulum is given by $T = 2\pi\sqrt{\frac{I}{mgl}}$ where m = mass of the pendulum, I = moment of inertia about the axis of suspension, l = distance of centre of mass of body from the axis of suspension. The change in time period when temperature changes by $\Delta\theta$ is given by $\frac{\alpha\Delta\theta}{n}T$, then the value of n is_______. [The coefficient of linear expansion of the material of pendulum is α]

2

52. Two spheres A and B of the same mass m and the same radius are placed on a rough horizontal surface. A is a uniform hollow sphere and B is uniform solid sphere. A and B can roll without sliding on the floor. They are also tied centrally to a light spring of spring constant k as shown in figure. A and B are released when the extension in spring is x_0. The amplitude of SHM of the sphere A is $\frac{n\,x_0}{46}$ then n is _______.



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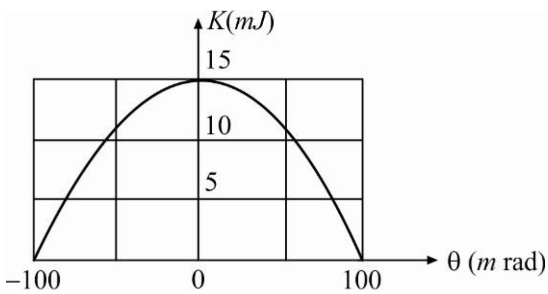
53. In a resonance tube experiment, a $1m$ long pipe is resonated with a tuning fork of frequency 340 Hz. Initially the pipe is completely filled with water. A small hole is drilled very close to the bottom and water is allowed to leak. If the radii of the pipe and hole are 2 cm and 1 mm respectively, then the time interval between the occurrence of first two resonances is ________ seconds. Neglect end correction. (Take speed of sound = $340\,m/s$, $g = 10\ m/s^2$, $\sqrt{5} = 2.2$, $\sqrt{3} = 1.73$)

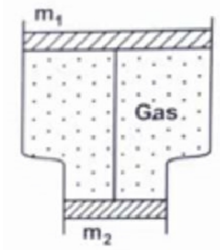
54. Figure shows the kinetic energy K of a simple pendulum versus its angle θ from the vertical. The pendulum bob has mass 0.2 kg. If the length of the pendulum is equal to $\frac{x}{2}m$, then value of x is _______.

$(g = 10\,\text{m/s}^2)$



55. Three rods A, B and C having identical shape and size, are hinged together at ends to form an equilateral triangle. Rods A and B are made of same material having coefficient of linear thermal expansion α_1 while that of material of rod C is α_2. By what temperature should the system of rods be heated to increase the angle opposite to rod C by $\Delta\theta$.

56. A smooth vertical tube having two different cross sections is open from both the ends but closed by two sliding pistons as shown in figure. Pistons are tied with an inextensible string. One mole of an ideal gas is enclosed between the pistons. The difference in cross-sectional areas of pistons is ΔS and masses of pistons are m_1 and m_2. Find the temperature by which tube is raised so that pistons will be displaced by a length ℓ. Take atmospheric pressure to be P_0.



57. Figure shows two coherent sources S_1 and S_2 which emit sound of wavelength λ in phase. The separation between the sources is 3λ. A circular wire of large radius is placed in such a way that S_1S_2 lies in its plane and the middle point of S_1S_2 is at the centre of the wire. Find the angular positions θ on the wire for which constructive interference takes place.

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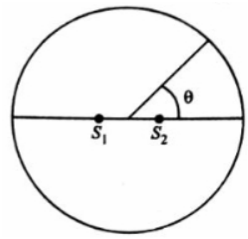

58. A soap bubble of radius r and surface tension T is given some charge and as a result of which its potential becomes V volt. Show that the new radius R of soap bubble is given by the equation

$$P_0(R^3-r^3)+4T(R^2-r^2)-\frac{\epsilon_0 V^2R}{2}=0$$

where P_0 is the atmospheric pressure.

59. A cubical container of side a and wall thickness $x(x<<a)$ is suspended in air and filled with n moles of a diatomic gas (adiabatic exponent $=\gamma$) in a room where room temperature is T_0. If at $t=0$ gas temperature is $T_1(T_1>T_0)$, find the temperature as a function of time t. Assume the heat is conducted through all the walls of container.

60. A bubble of radius r, containing a diatomic ideal gas, has the soap film of thickness h and is placed in vacuum. The soap film has surface tension σ and density ρ.

(i) Find the molar heat capacity of the gas in the bubble for such a process when the gas is heated so slowly that the bubble remains in a mechanical equilibrium.

(ii) Find the frequency ω of the small radial oscillations of the bubble under the assumption that the heat capacity of the soap film is much greater than the heat capacity of the gas in the bubble. Assume that the thermal equilibrium inside the bubble is reached much faster than the period of oscillations.